Dublin City University

SEMESTER TWO EXAMINATION 2018 – 2019

Module	**MS200 – Linear Algebra**
Programme(s)	B. Sc. in Computer Applications B. Sc. in Science Education B. Sc. in Physical Education with Mathematics B. Sc in Environmental Science and Health
Year of Study	2
Examiner(s)	Prof. D. Wraith Dr. M. Clancy (Ext. 5774)
Time Allowed	**2** hours
Instructions	**Answer ALL Questions.**
Requirements	Log tables to be supplied. Approved calculators may be used.

PLEASE DO NOT TURN OVER THIS PAGE UNTIL INSTRUCTED TO DO SO.
The use of programmable or text-storing calculators is expressly forbidden.

QUESTION 1: Throughout this question $\boldsymbol{u}$ and $\boldsymbol{v}$ will be the following vectors in $\mathbb{R}^3$:

$$\boldsymbol{u} = \begin{bmatrix} 2 \\ 1 \\ -1 \end{bmatrix} \quad \text{and} \quad \boldsymbol{v} = \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix}.$$

Do the following:

(a) Calculate the vector $(-2\,\boldsymbol{u} + 5\,\boldsymbol{v})$.

[3 marks]

(b) Calculate the length of the vector $\boldsymbol{u}$.

[3 marks]

(c) Calculate the **angle** θ (measured in degrees) between the vectors $\boldsymbol{u}$ and $\boldsymbol{v}$.

[4 marks]

QUESTION 2

Throughout this question $\boldsymbol{u}_1,\ \boldsymbol{u}_2,\ \boldsymbol{u}_3$ and $\boldsymbol{v}$ will be the following vectors in $\mathbb{R}^4$:

$$\boldsymbol{u}_1 = \frac{1}{\sqrt{2}}\begin{bmatrix} -1 \\ 0 \\ 1 \\ 0 \end{bmatrix}, \quad \boldsymbol{u}_2 = \frac{1}{\sqrt{2}}\begin{bmatrix} 1 \\ 0 \\ 1 \\ 0 \end{bmatrix}, \quad \boldsymbol{u}_3 = \frac{1}{\sqrt{2}}\begin{bmatrix} 0 \\ 1 \\ 0 \\ 1 \end{bmatrix} \quad \text{and} \quad \boldsymbol{v} = \begin{bmatrix} 2 \\ 1 \\ -3 \\ -1 \end{bmatrix}.$$

Let $U \subset \mathbb{R}^4$ be the subspace of $\mathbb{R}^4$ that is spanned by $\{\boldsymbol{u}_1,\ \boldsymbol{u}_2,\ \boldsymbol{u}_3\}$. Given that $\{\boldsymbol{u}_1,\ \boldsymbol{u}_2,\ \boldsymbol{u}_3\}$ is an **orthonormal set of vectors** (relative to the usual inner product on $\mathbb{R}^4$) do the following:

(a) Calculate the **orthogonal projection** (denoted by $P_U\boldsymbol{v}$) of $\boldsymbol{v}$ onto the subspace U.

[6 marks]

(b) Find a **non-zero** vector $\boldsymbol{w} \in \mathbb{R}^4$ that is perpendicular to the subspace U.

[4 marks]

QUESTION 3: Throughout this question the plane, P, in $\mathbb{R}^3$ will be the solution set of the equation $2x + y - 2z = 0$ and the vector $\boldsymbol{v}$ in $\mathbb{R}^3$ will be:

$$\boldsymbol{v} = \begin{bmatrix} 1 \\ -3 \\ 1 \end{bmatrix}.$$

This vector $\boldsymbol{v}$ can be written in the form $\boldsymbol{v} = \boldsymbol{v}_P + \boldsymbol{v}_n$, where $\boldsymbol{v}_P$ is the component of $\boldsymbol{v}$ that lies on the plane P and $\boldsymbol{v}_n$ is the component of $\boldsymbol{v}$ that is normal to the plane P. Do the following:

(a) Calculate the vector $\boldsymbol{v}_n$.

[5 marks]

(b) Suppose that this plane P is in fact a mirror. If a ray of light that is travelling in the direction of the vector $\boldsymbol{v}$ is reflected off P find a (non-zero) vector $\boldsymbol{w}$ that points in the direction of the reflected ray.

Hint: Draw a schematic diagram to illustrate the plane P together with the vectors $\boldsymbol{v}$, $\boldsymbol{v}_P$, $\boldsymbol{v}_n$ and $\boldsymbol{w}$ and use the result that you have obtained in part (a).

[5 marks]

QUESTION 4: If A, B and $\boldsymbol{x}$ are given by:

$$A = \begin{bmatrix} 3 & -2 & 0 \\ -2 & 0 & 1 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & -2 \\ 0 & 1 \\ -2 & 0 \end{bmatrix}, \quad \boldsymbol{x} = \begin{bmatrix} 1 \\ 3 \\ -2 \end{bmatrix},$$

do the following:

(a) Calculate $A\boldsymbol{x}$, AB and the **transpose**, A^T, of A.

[6 marks]

(b) If C is an $(m \times p)$ - matrix and if D is a $(p \times n)$ - matrix, use the formula

$$(CD)_{ij} = \sum_{k=1}^{p} C_{ik} D_{kj}$$

for calculating the ij^{th} entry of the product CD to show that $(CD)^T = D^T C^T$.

Hint : Begin with $[(CD)^T]_{ij} = (CD)_{ji}$.

[4 marks]

QUESTION 5

(a) Write down the augmented matrix corresponding to the system of linear equations:

$$\boxed{\begin{array}{rcrcrcrcr} w & + & 2x & - & y & + & 5z & = & -3 \\ & - & 4x & + & y & - & z & = & 1 \\ 2w & + & 4x & - & 5y & - & 2z & = & 7 \end{array}}$$

[2 marks]

(b) For the remainder of this question the variables u, v, w, x, y, and z will satisfy a system of linear equations whose **augmented matrix** is $[\,A\,|\,\boldsymbol{b}\,]$. If the **reduced row echelon form** of $[\,A\,|\,\boldsymbol{b}\,]$ is:

$$\left[\begin{array}{cccccc|c} 1 & 2 & 0 & -2 & 0 & -3 & -2 \\ 0 & 0 & 1 & 3 & 0 & 4 & 6 \\ 0 & 0 & 0 & 0 & 1 & -5 & 3 \end{array}\right],$$

do the following:

(i) Parametrize (as a subset of $\mathbb{R}^6$) the solution space of this system of equations in the variables u, v, w, x, y, and z.

[5 marks]

(ii) Determine a basis for the **kernel** of the matrix A.

[3 marks]

QUESTION 6: Throughout this question the vector $\boldsymbol{u} \in \mathbb{R}^3$, the vectors $\boldsymbol{v}_1, \boldsymbol{v}_2 \in \mathbb{R}^2$ and the matrix A will be as follows:

$$\boldsymbol{u} = \begin{bmatrix} 1 \\ -1 \\ 2 \end{bmatrix}, \quad \boldsymbol{v}_1 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}, \quad \boldsymbol{v}_2 = \begin{bmatrix} 1 \\ 3 \end{bmatrix} \quad \text{and} \quad A = \begin{bmatrix} 1 & 3 \\ 2 & -4 \\ -1 & 0 \end{bmatrix}.$$

Answer the following, **making sure that in each case the logic of your argument is explained clearly**:

(a) Is the vector $\boldsymbol{u}$ in the **image** of A?

[4 marks]

(b) Is $\{A\boldsymbol{v}_1, A\boldsymbol{v}_2\}$ a **linearly independent** set of vectors?

[3 marks]

(c) What is the dimension of the **kernel** of A?

[3 marks]

QUESTION 7

Throughout this question $\{\boldsymbol{e}_1,\, \boldsymbol{e}_2,\, \boldsymbol{e}_3\}$ will denote the usual basis for $\mathbb{R}^3$. If $L : \mathbb{R}^3 \to \mathbb{R}^3$ is the linear map that is specified by:

$$L\boldsymbol{e}_1 = \frac{1}{2}(\boldsymbol{e}_1 + \boldsymbol{e}_2 + \sqrt{2}\boldsymbol{e}_3), \quad L\boldsymbol{e}_2 = \frac{\sqrt{2}}{2}(\boldsymbol{e}_1 - \boldsymbol{e}_2) \;\text{ and }\; L\boldsymbol{e}_3 = \frac{1}{2}(\boldsymbol{e}_1 + \boldsymbol{e}_2 - \sqrt{2}\boldsymbol{e}_3),$$

do the following:

(a) Calculate the matrix, M_L, that represents the map L relative to $\{\boldsymbol{e}_1,\, \boldsymbol{e}_2,\, \boldsymbol{e}_3\}$.

[3 marks]

(b) Perform the appropriate calculations to show that the map L is indeed a rotation (about the origin) of $\mathbb{R}^3$.

[3 marks]

(c) Find the axis of rotation of the map L.

[4 marks]

QUESTION 8

If the matrix $A = \begin{bmatrix} -3 & 0 & 0 \\ -4 & -7 & 8 \\ -4 & -4 & 5 \end{bmatrix}$, do the following:

(a) Find the **characteristic polynomial** of A and, hence, find all of the eigenvalues of A.

[5 marks]

(b) Parametrize (as a subset of $\mathbb{R}^3$) the **eigenspace** of A that corresponds to the eigenvalue $\lambda = -3$. Is it possible to diagonalize A? Justify your answer.

[5 marks]

QUESTION 9: Throughout this question $\boldsymbol{u}$, $\boldsymbol{v}$, $\boldsymbol{w}$ will be the following vectors:

$$\boldsymbol{u} = \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix}, \quad \boldsymbol{v} = \begin{bmatrix} -1 \\ -1 \\ 1 \end{bmatrix} \text{ and } \boldsymbol{w} = \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix}.$$

If a 3×3 matrix A is such that

$$A\boldsymbol{u} = -\boldsymbol{u}, \quad A\boldsymbol{v} = \boldsymbol{v} \text{ and } A\boldsymbol{w} = 2\boldsymbol{w},$$

do the following:

(a) Find a 3×3 invertible real matrix P and a 3×3 real **diagonal matrix** Λ such that

$$P^{-1}AP = \Lambda.$$

[5 marks]

(b) Determine, with justification, whether or not the equation $A\boldsymbol{x} = \boldsymbol{0}$ has a **non-zero** solution $\boldsymbol{x} \in \mathbb{R}^3$.

[5 marks]

QUESTION 10

(a) Write down the 3×3 elementary matrix E, such that the matrix EA is obtained from the matrix A when:

the second row of A is is interchanged with the **first row** of A.

What is the **inverse** of this 3×3 elementary matrix E?

[4 marks]

(b) Let $A = \begin{bmatrix} 0 & -1 & 2 \\ -1 & -1 & 1 \\ 2 & 1 & 1 \end{bmatrix}$ and let $\tilde{A} = \begin{bmatrix} -1 & -1 & 1 \\ 0 & -1 & 2 \\ 2 & 1 & 1 \end{bmatrix}$, that is, $\tilde{A} =$ the matrix A with its first and second rows interchanged. Given that

$$A^{-1} = \begin{bmatrix} 2 & -3 & -1 \\ -3 & 4 & 2 \\ -1 & 2 & 1 \end{bmatrix},$$

determine $(\tilde{A})^{-1}$. **Hint:** This can be done by a method that is much simpler than that of Gauss-Jordan elimination!

[6 marks]